THE PARABOLA

CONICAL SECTIONS

prepared by:

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Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

40%

WEEKLY

60%

EXAM

PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

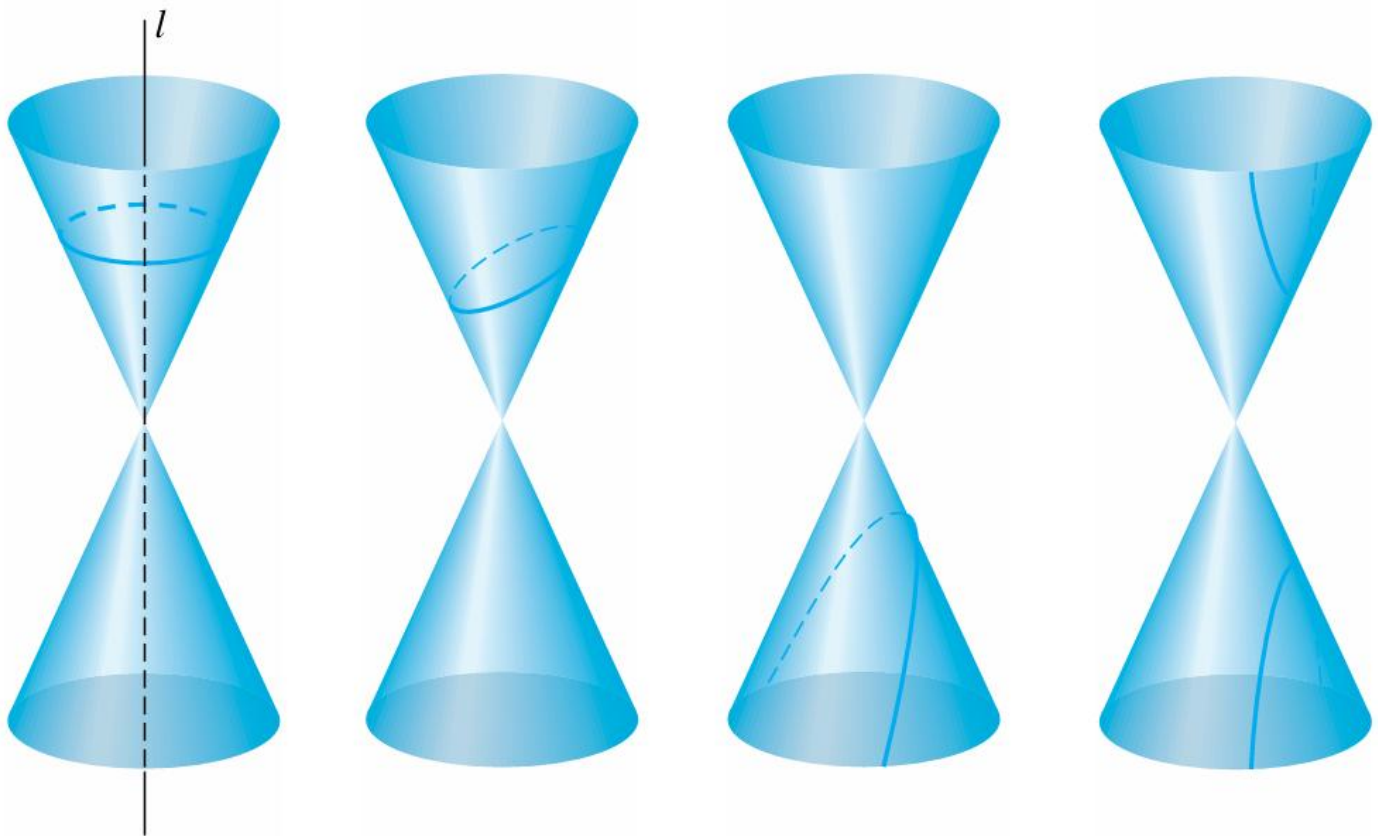
TOPIC OUTLINE

Parabola with Vertex at the Origin

Parabola with Vertex at $V(h, k)$

PARABOLA WITH VERTEX AT THE ORIGIN

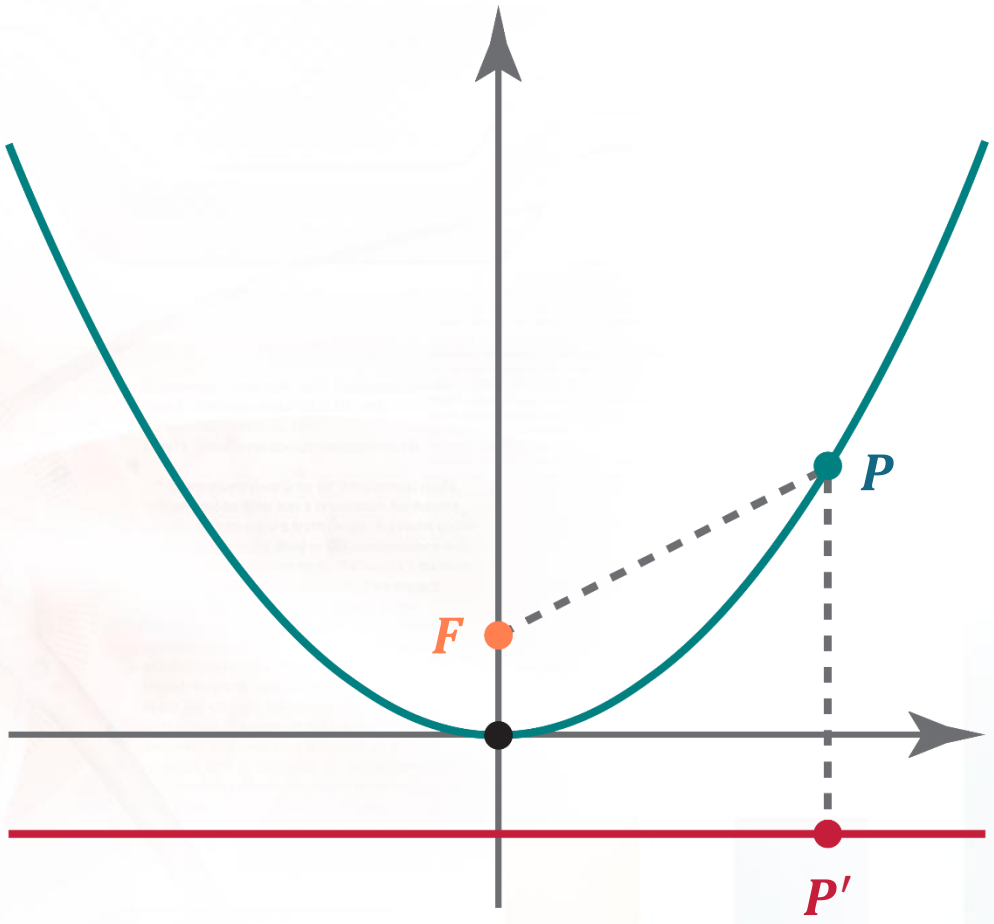
CONICAL SECTIONS



Circle Ellipse **Parabola** Hyperbola

DEFINITION

A **parabola** is the set of all points in a plane **equidistant** from a fixed point F (the **focus**) and a fixed line l (the **directrix**) that lie in the plane

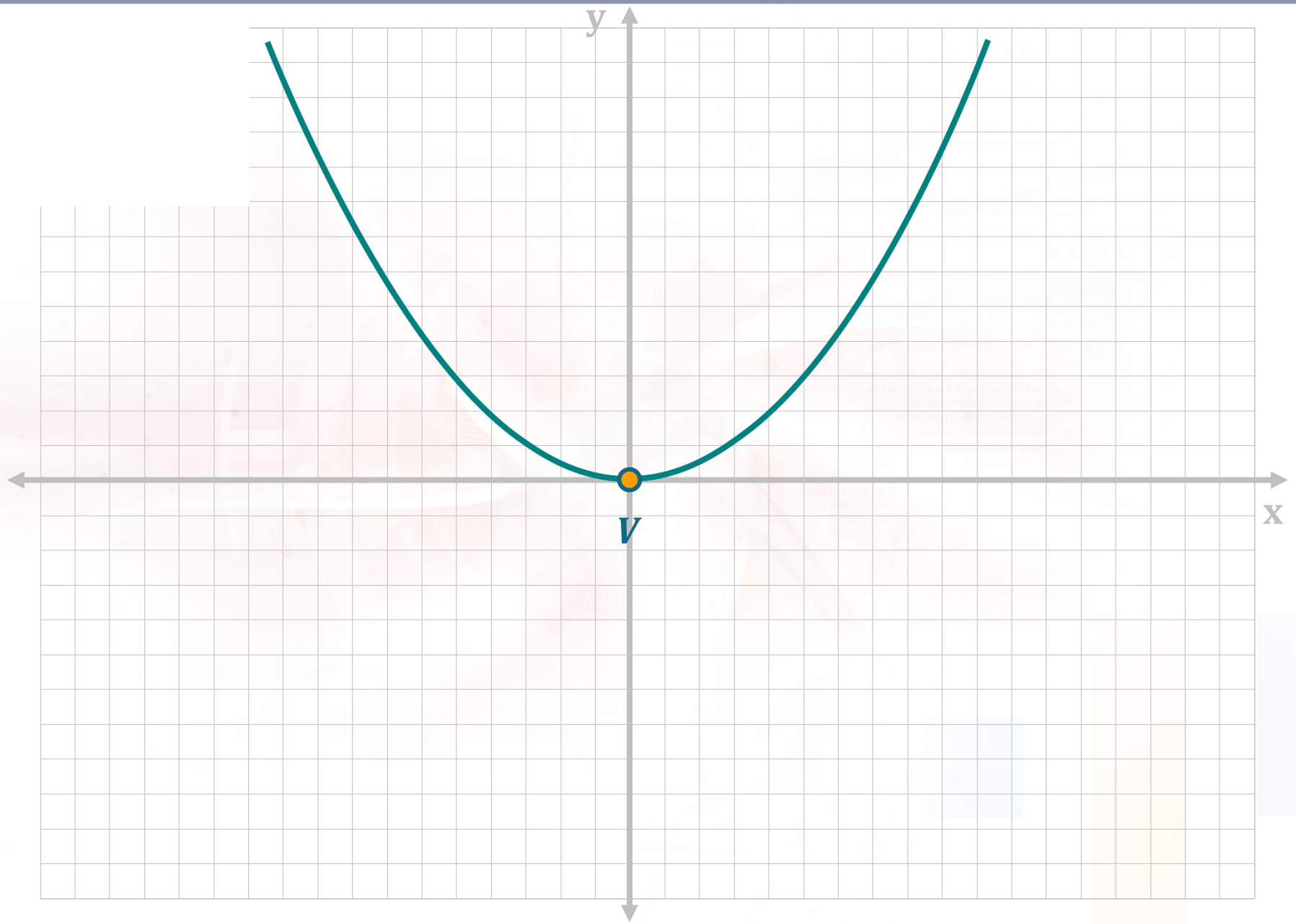


$$d(P, F) = d(P, P')$$

VERTEX AT THE ORIGIN

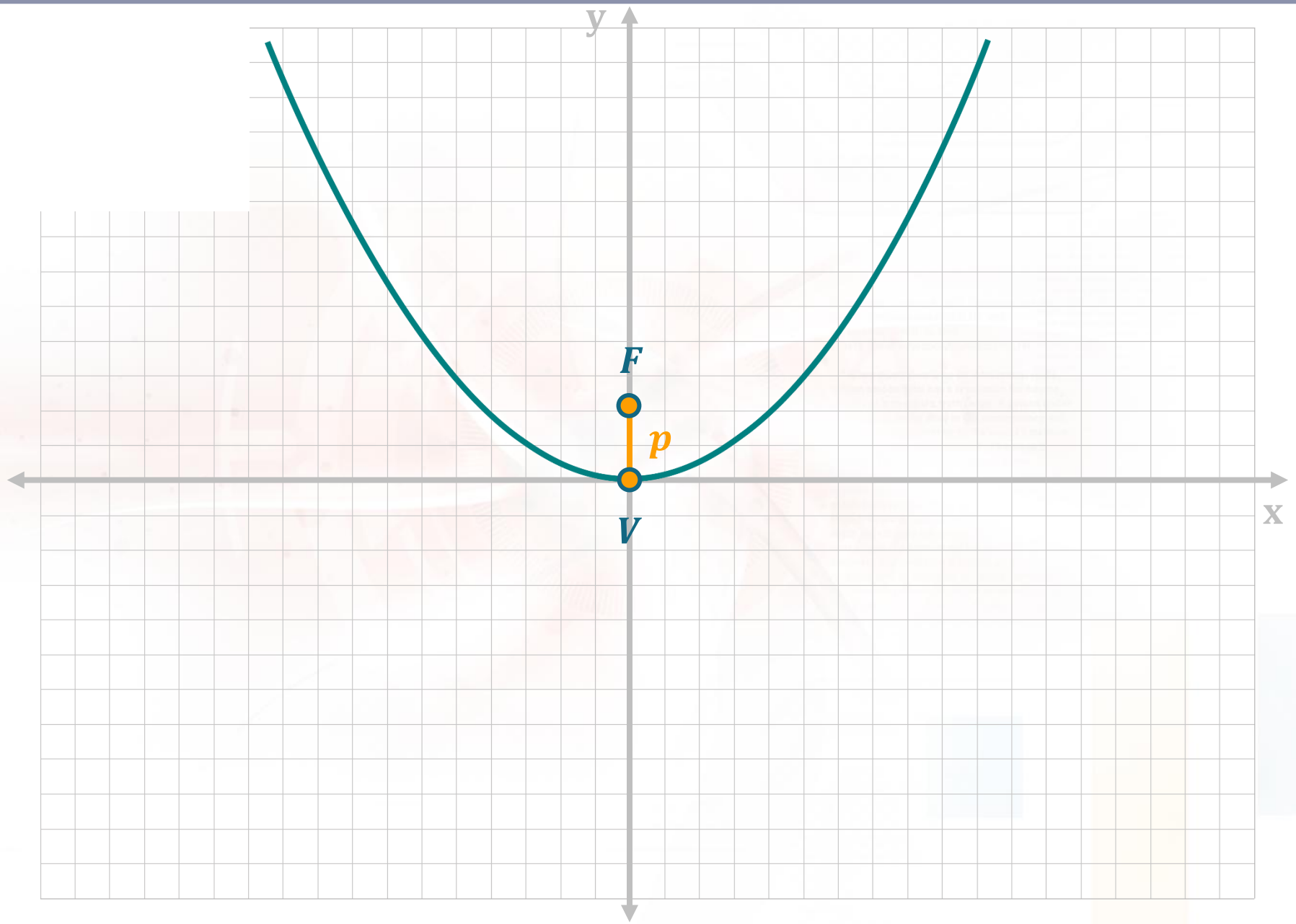
vertex

$V(0, 0)$



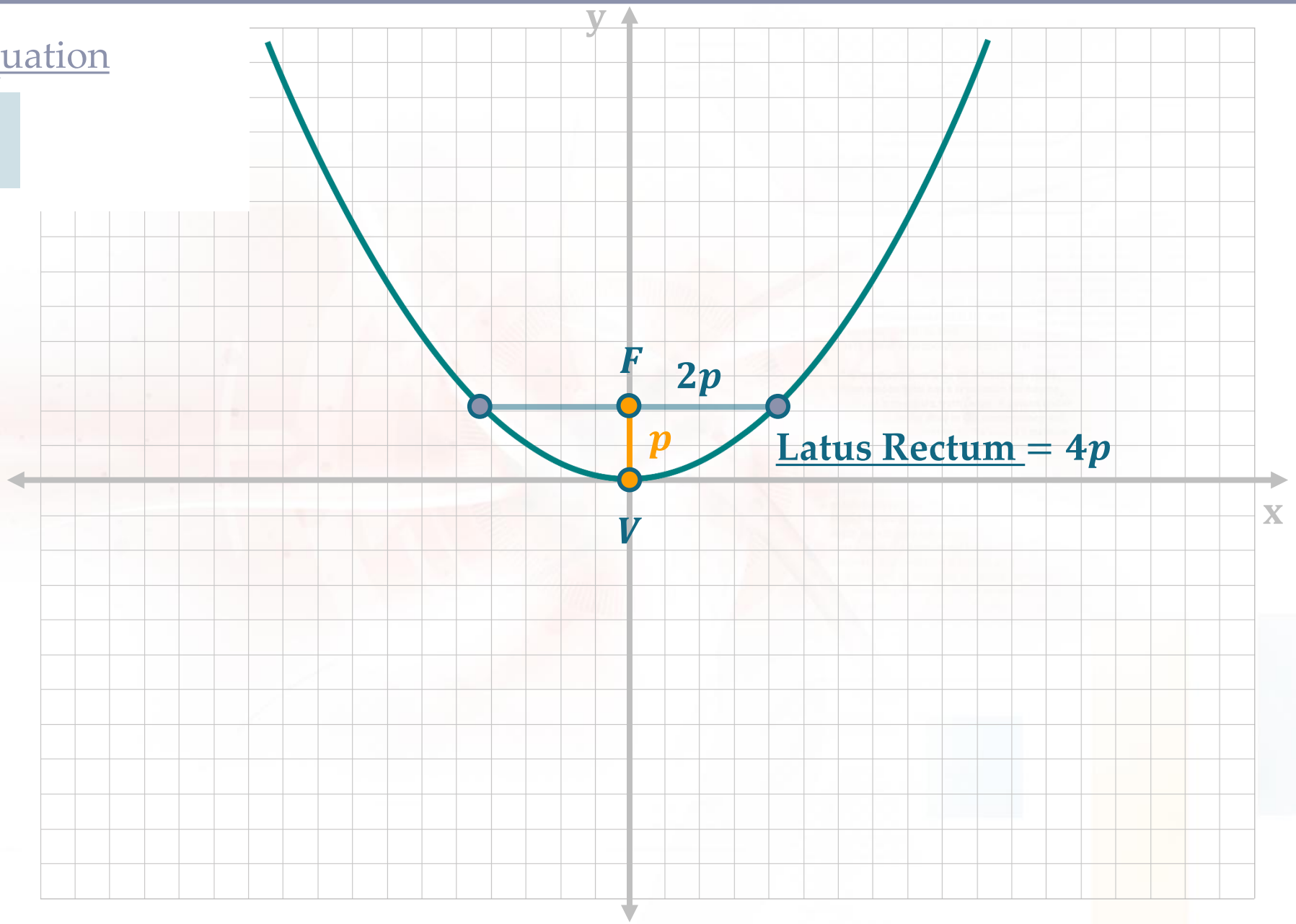
focus

$F(0, p)$



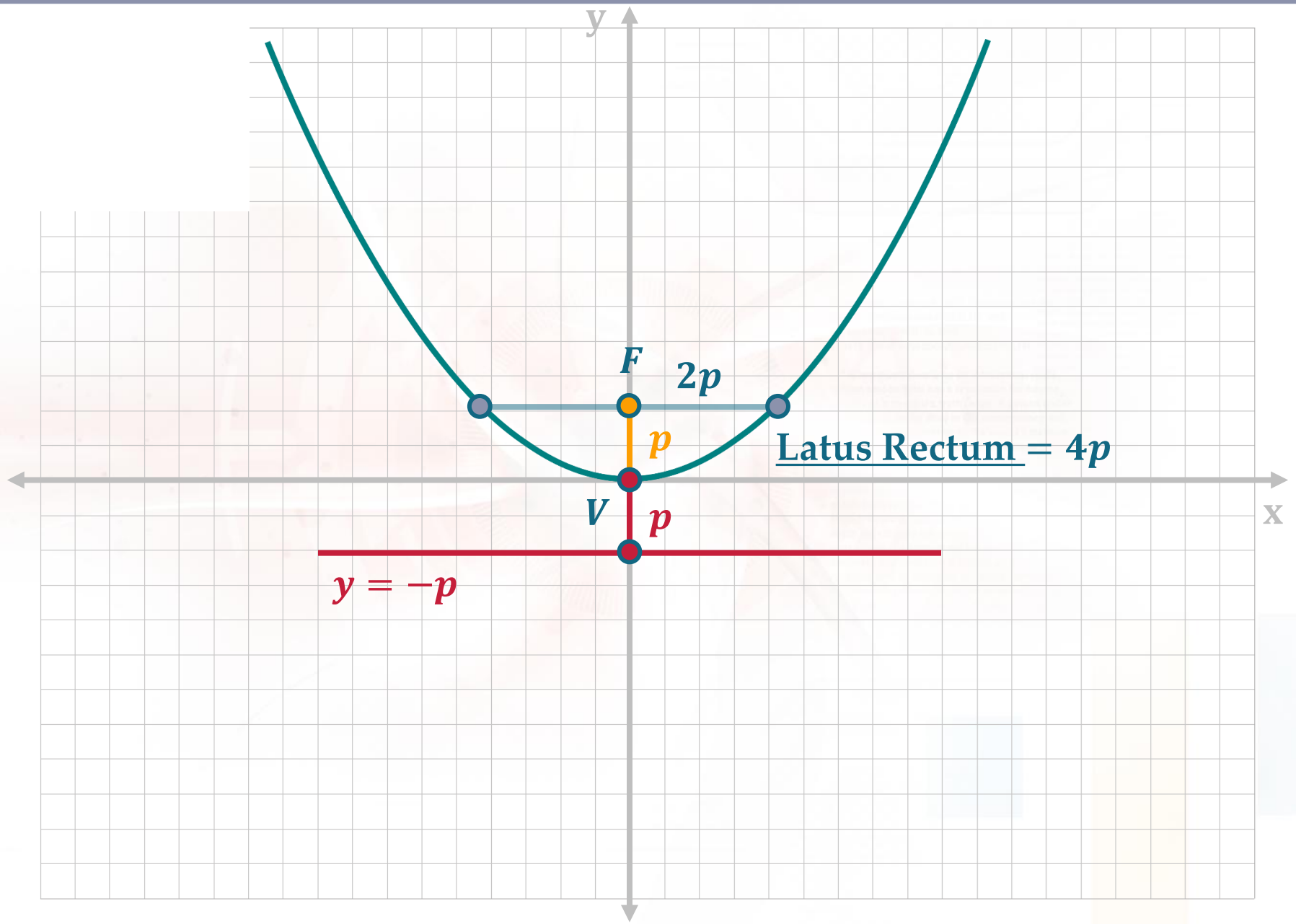
Standard Equation

$$x^2 = 4p\,y$$



directrix

$y = -p$



DIRECTRIX is along the $x - axis$

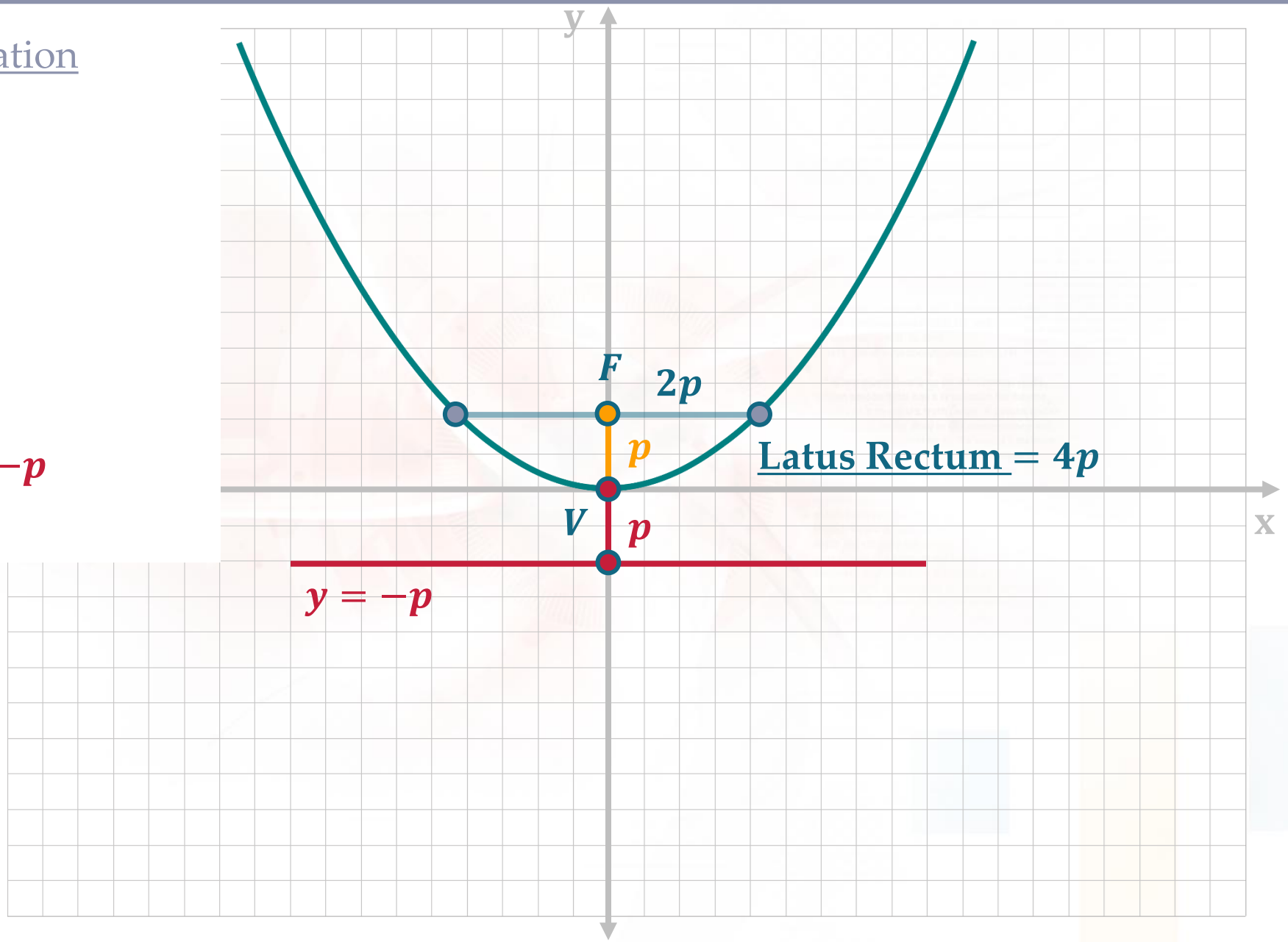
Standard Equation

$$x^2 = 4p\,y$$

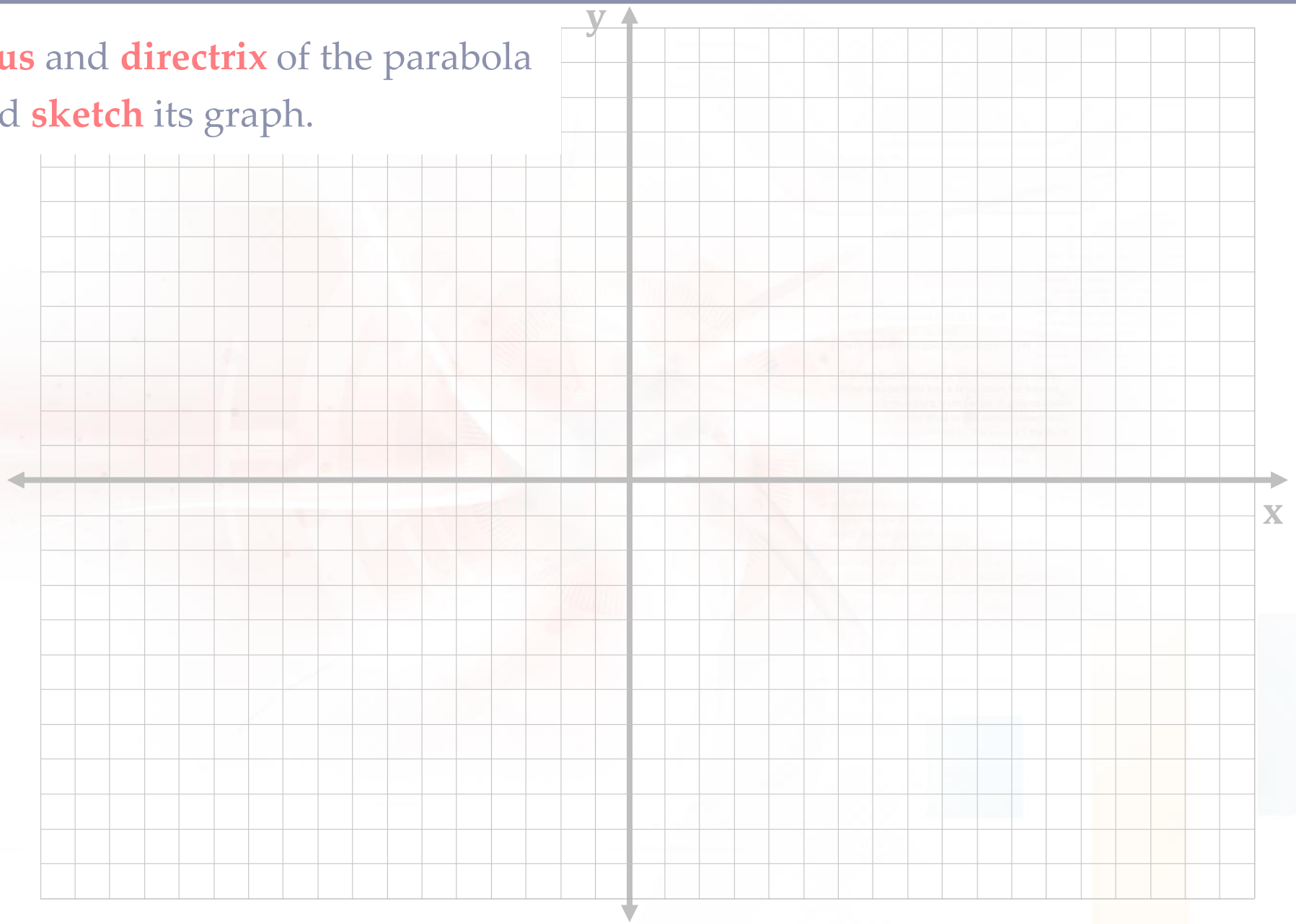
vertex, $V(0, 0)$

focus, $F(0, p)$

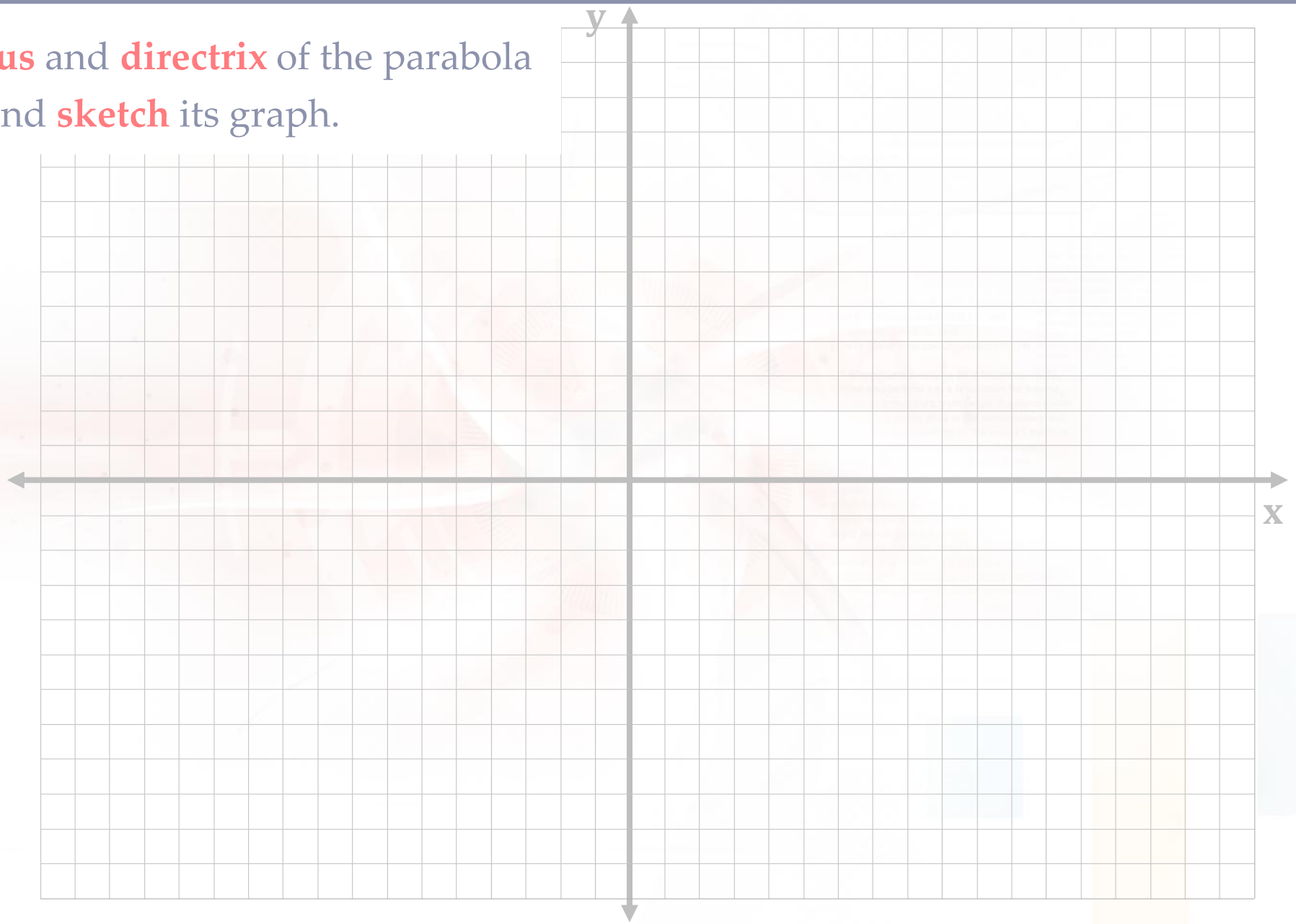
directrix, $y = -p$



Find the **focus** and **directrix** of the parabola $y = \frac{1}{6}x^2$, and **sketch** its graph.



Find the **focus** and **directrix** of the parabola $y = -\frac{1}{6}x^2$, and **sketch** its graph.



DIRECTRIX is along the $y - axis$

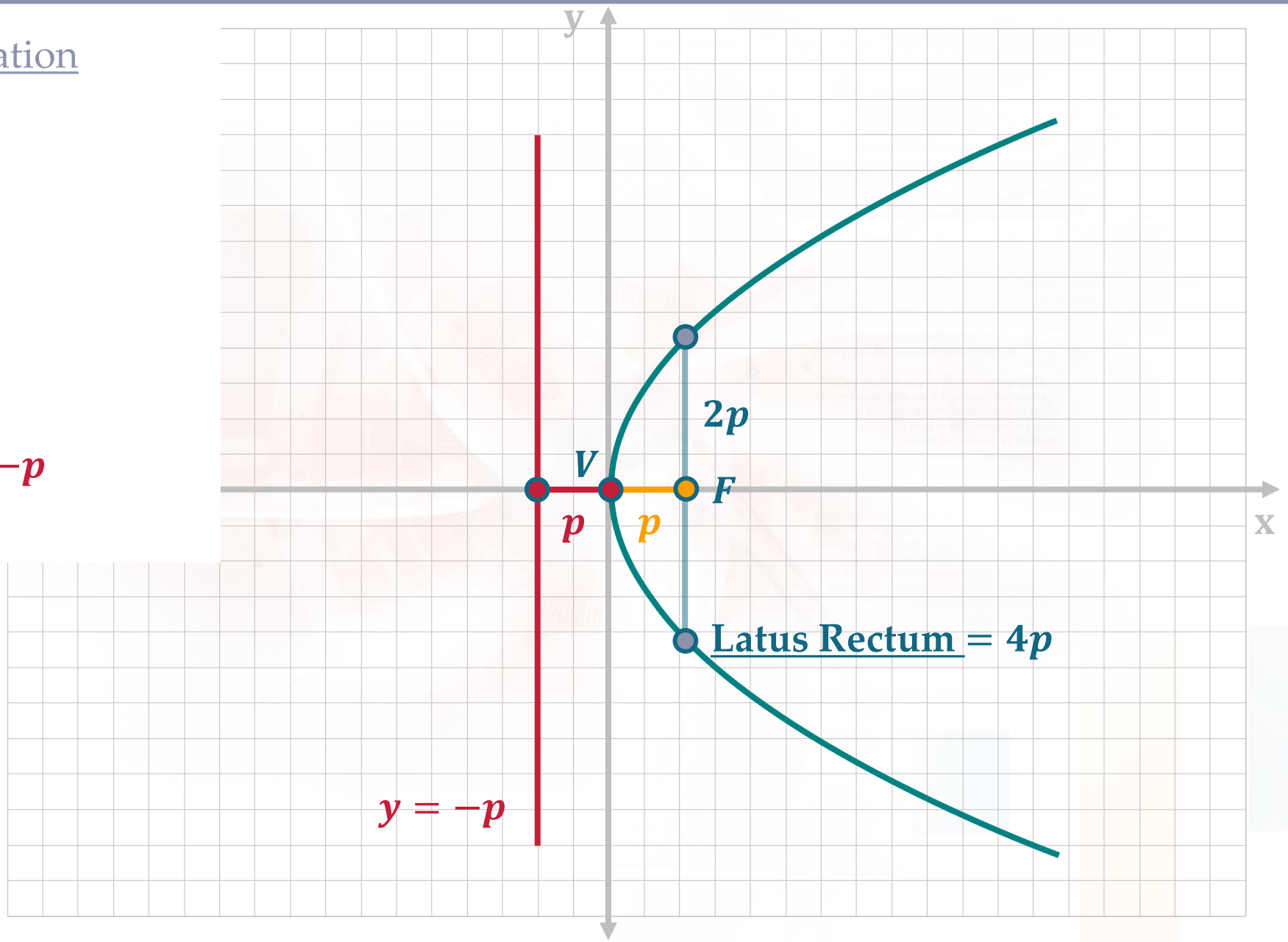
Standard Equation

$$y^2 = 4p\,x$$

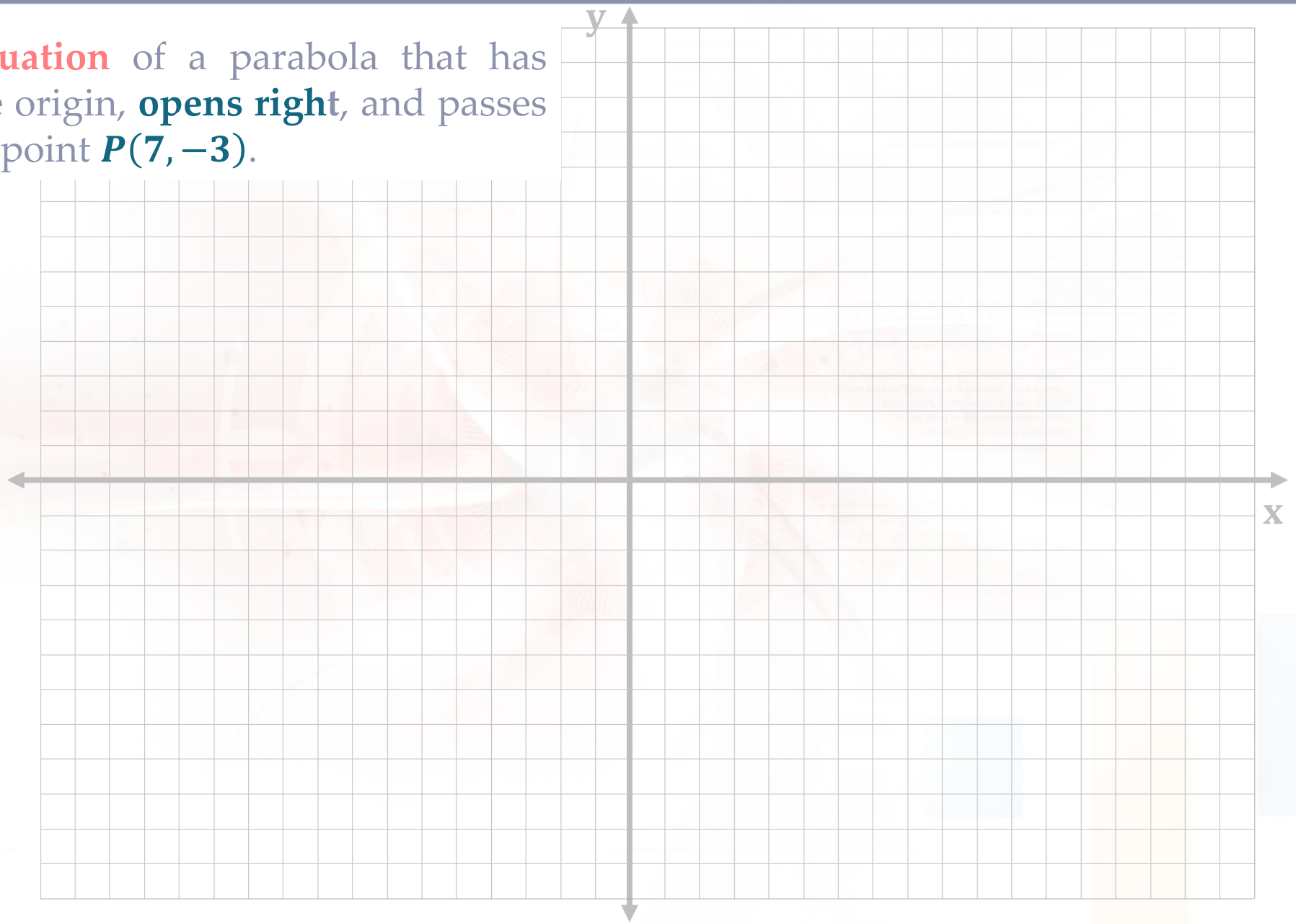
vertex, $V(0, 0)$

focus, $F(p, 0)$

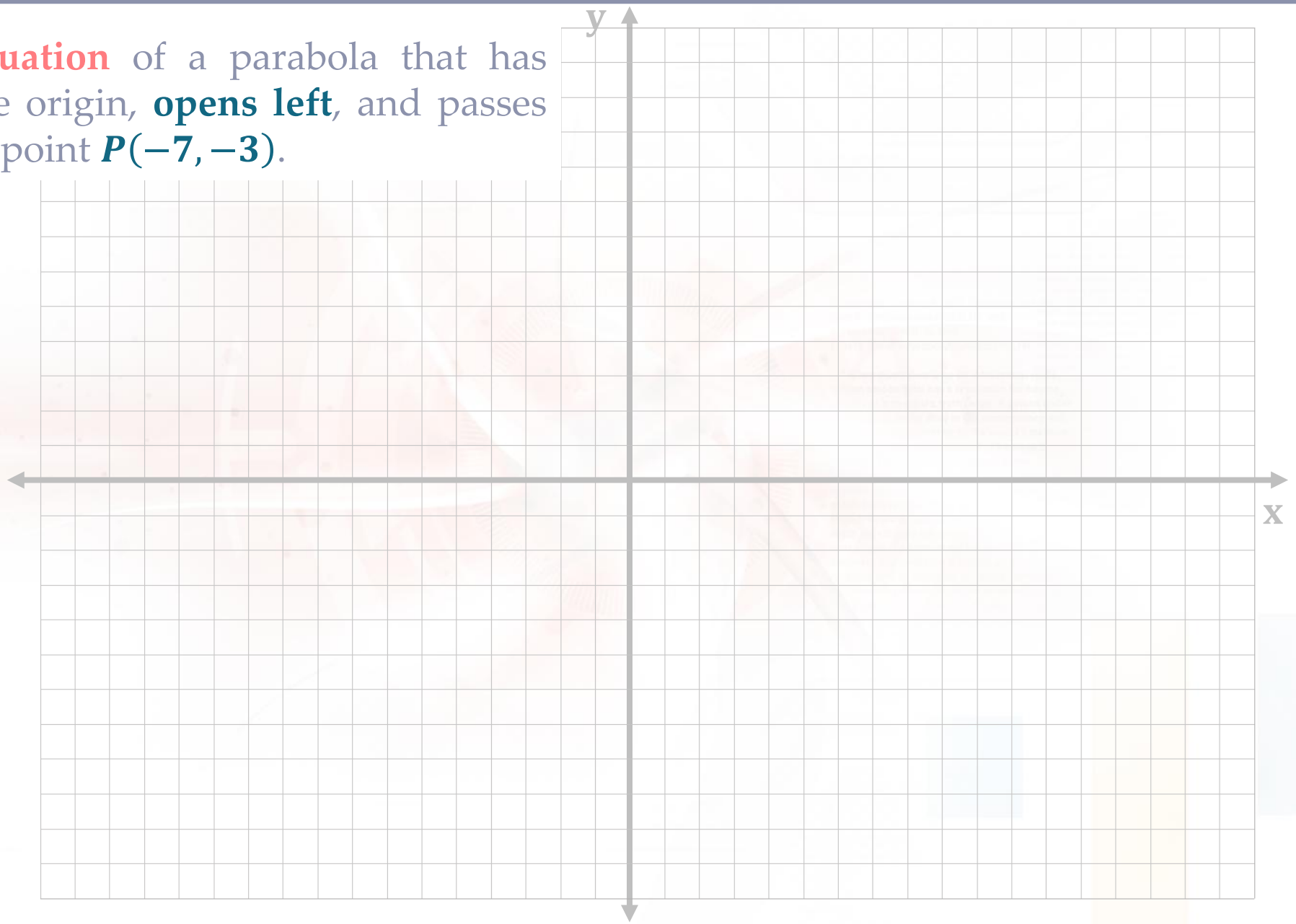
directrix, $x = -p$



Find an **equation** of a parabola that has vertex at the origin, **opens right**, and passes through the point $\boldsymbol{P(7,-3)}$.



Find an **equation** of a parabola that has vertex at the origin, **opens left**, and passes through the point $\boldsymbol{P(-7,-3)}$.

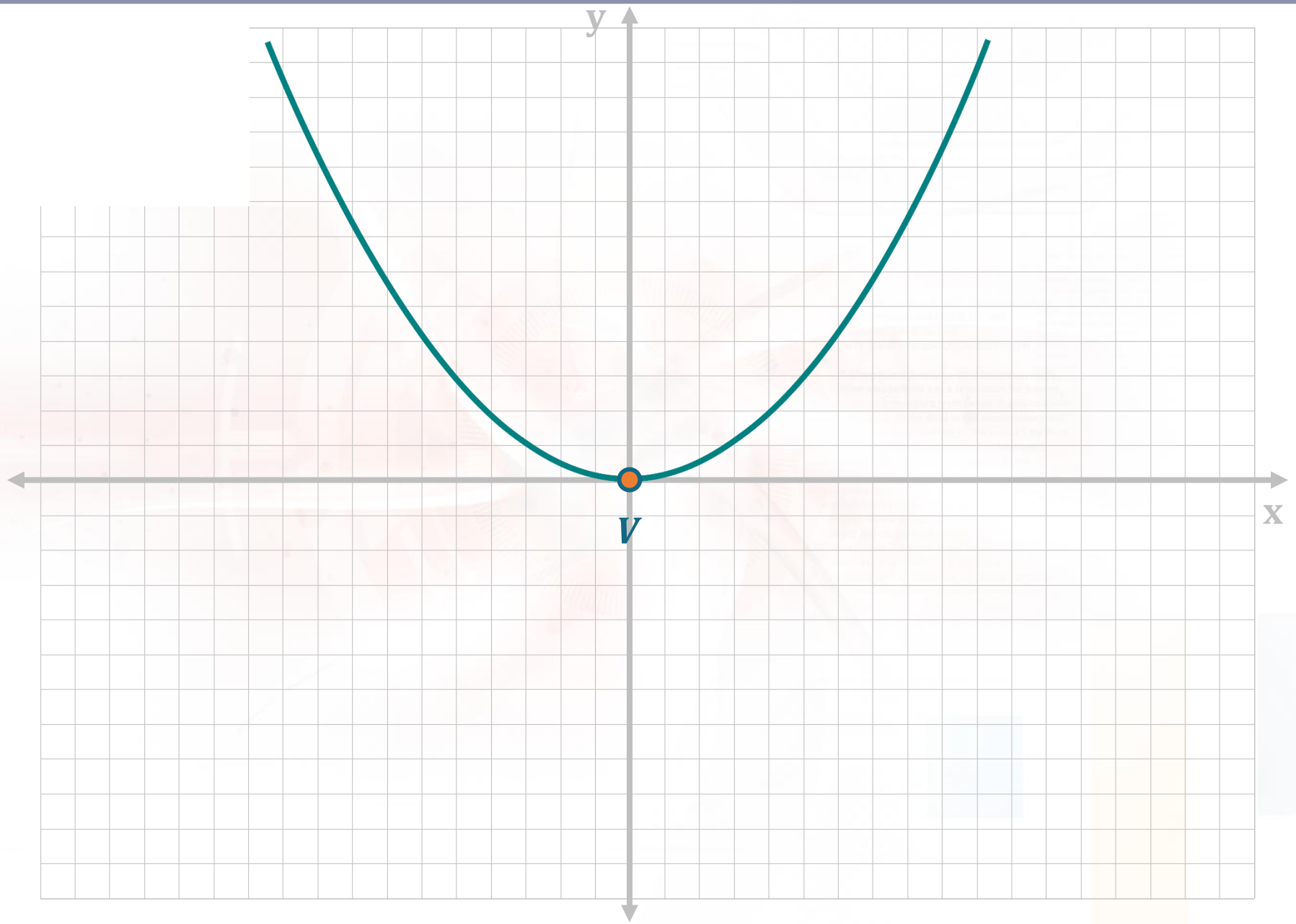


PARABOLA WITH VERTEX AT $V(h, k)$

VERTEX AT THE ORIGIN

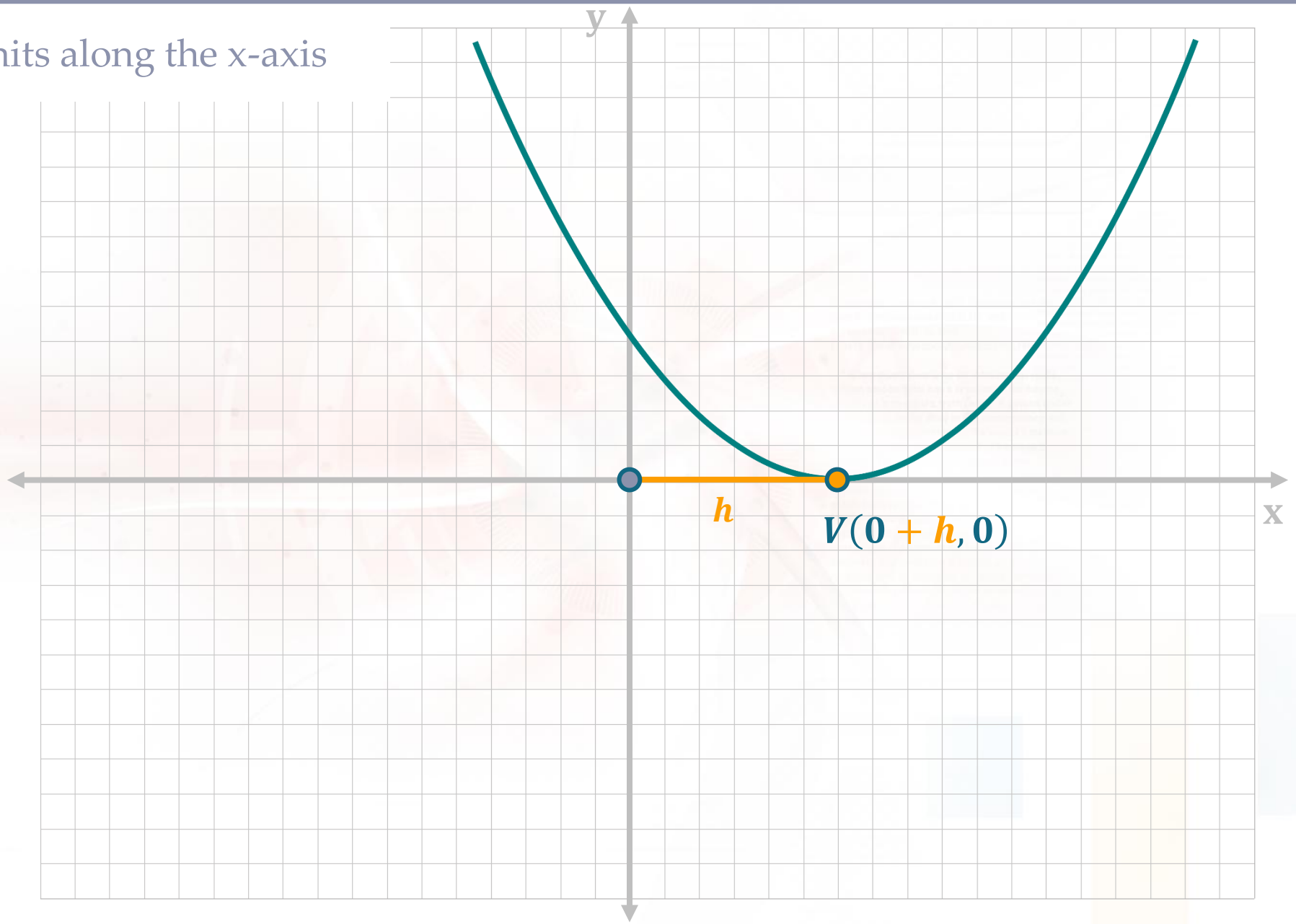
vertex

$V(0, 0)$



VERTEX AT $V(h, 0)$

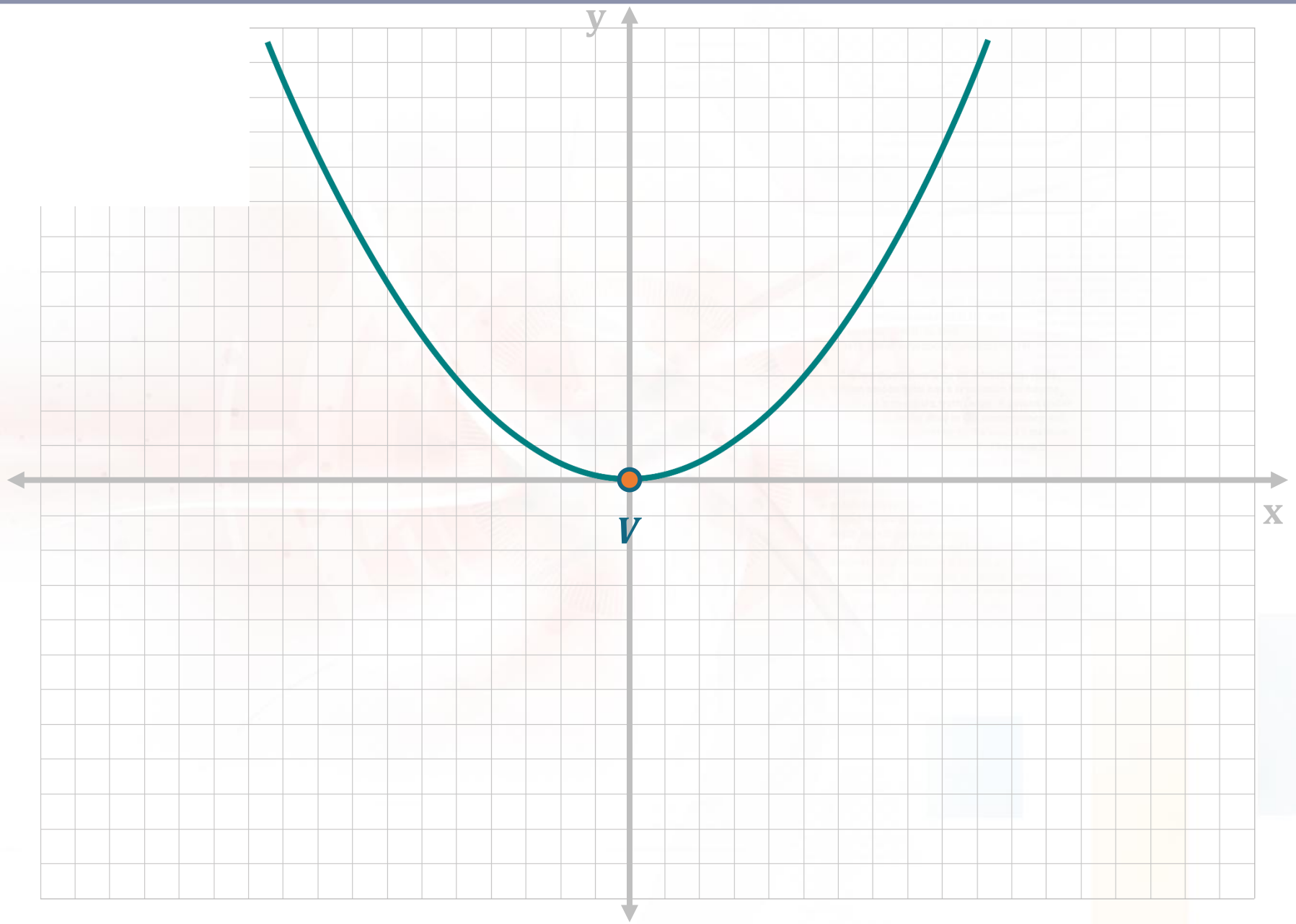
moving h units along the x-axis



VERTEX AT THE ORIGIN

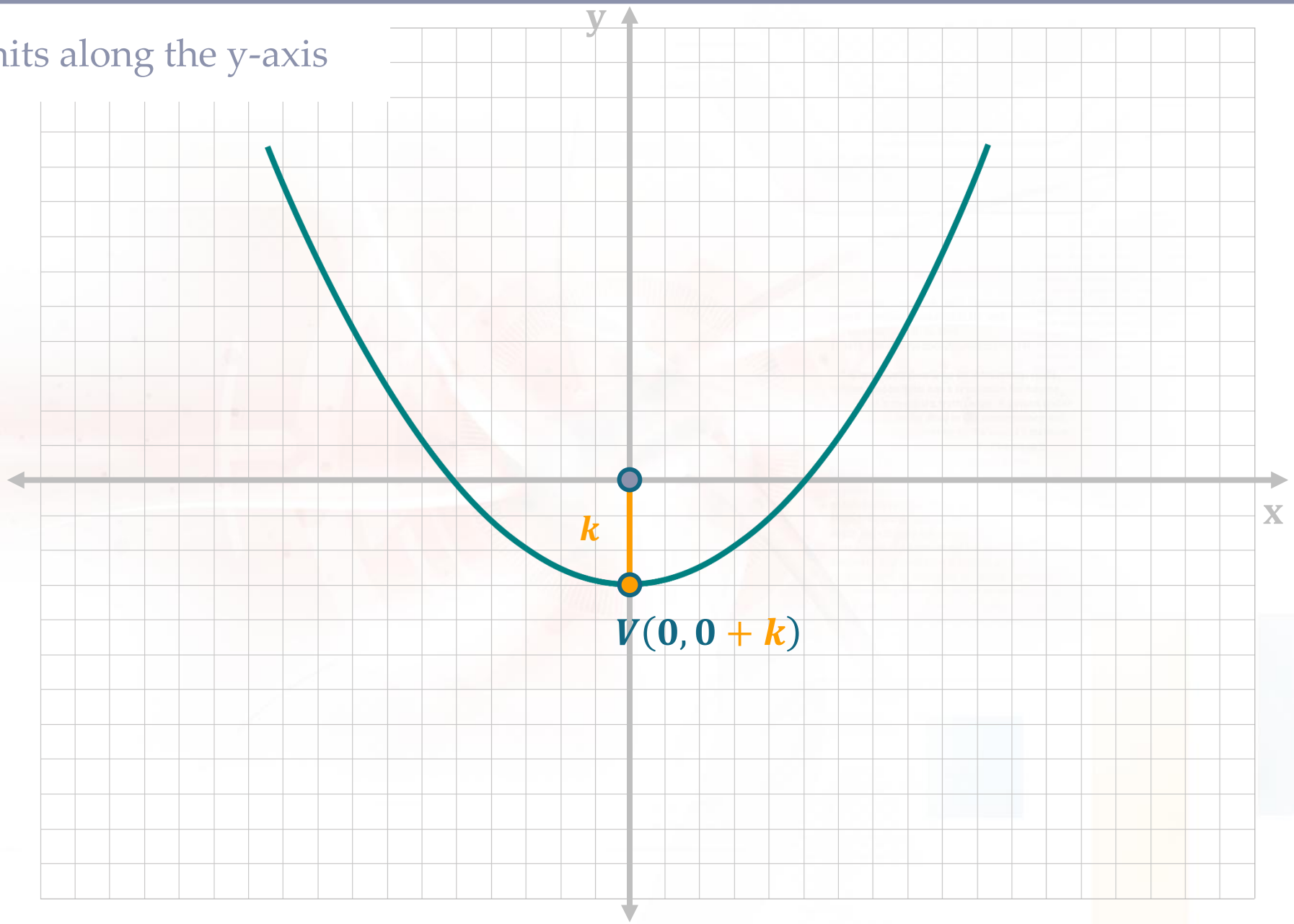
vertex

$V(0, 0)$



VERTEX AT $V(0, k)$

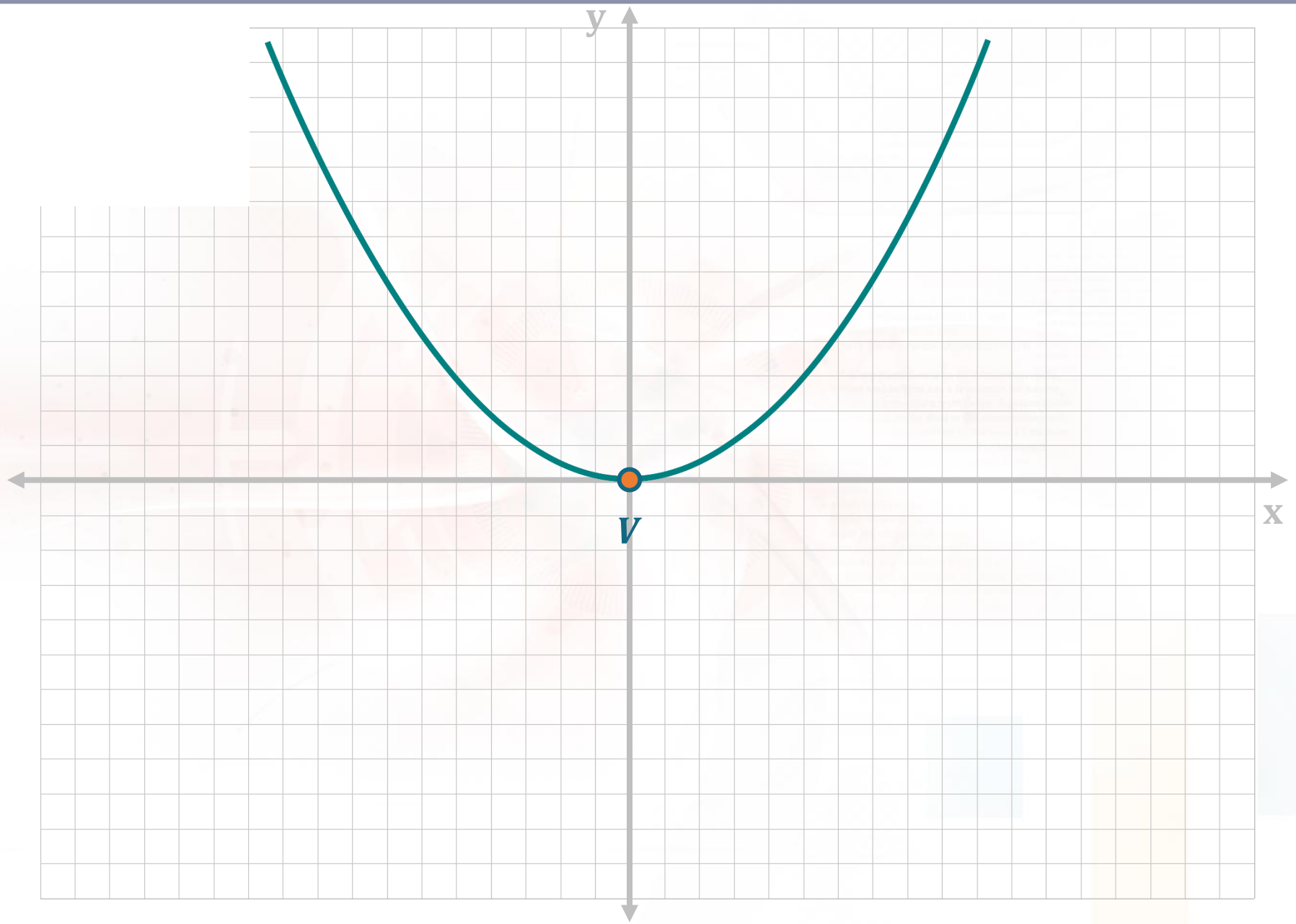
moving k units along the y-axis



VERTEX AT THE ORIGIN

vertex

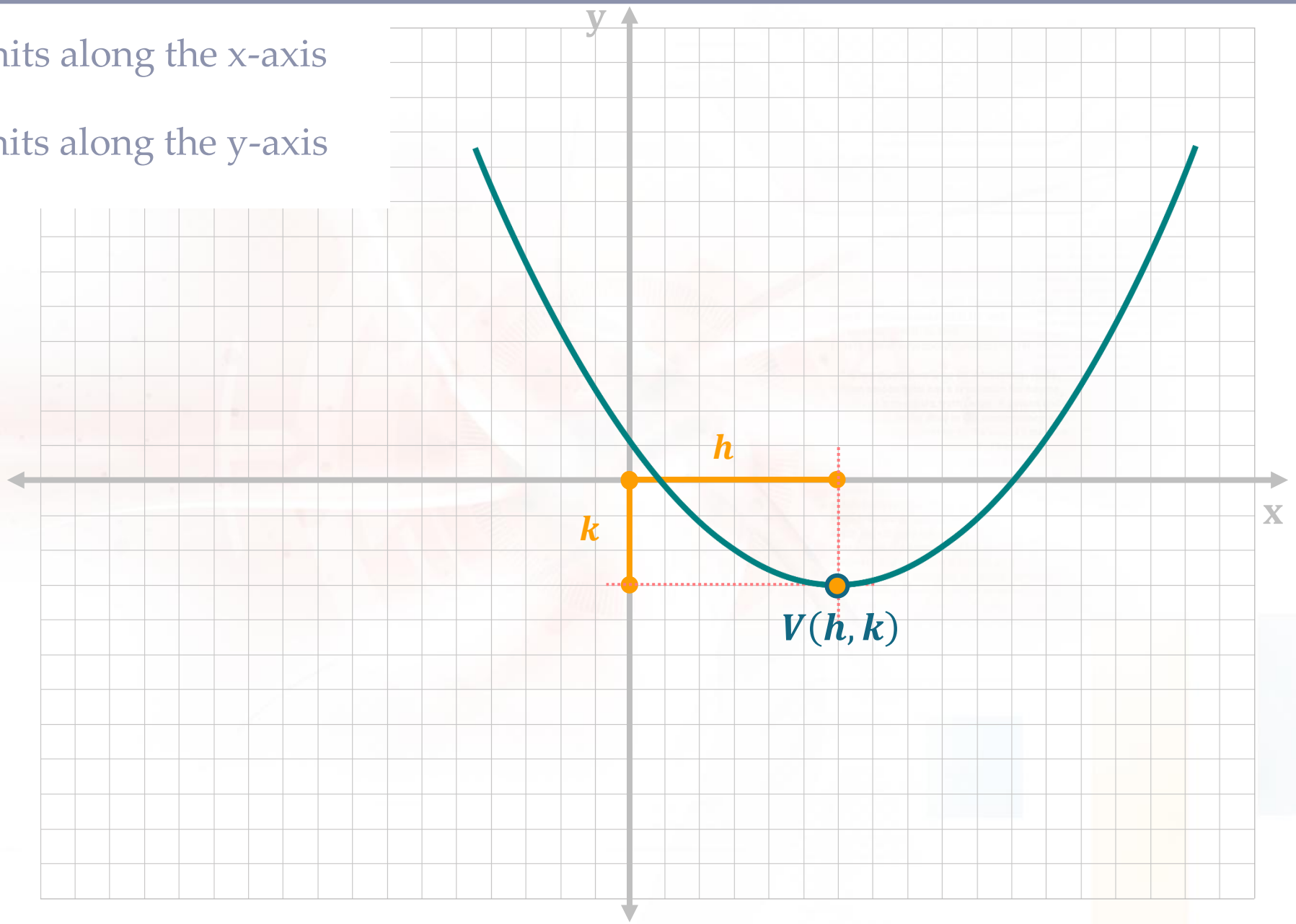
$V(0, 0)$



VERTEX AT $V(h, k)$

moving h units along the x-axis

moving k units along the y-axis



DIRECTRIX is along the $x - axis$

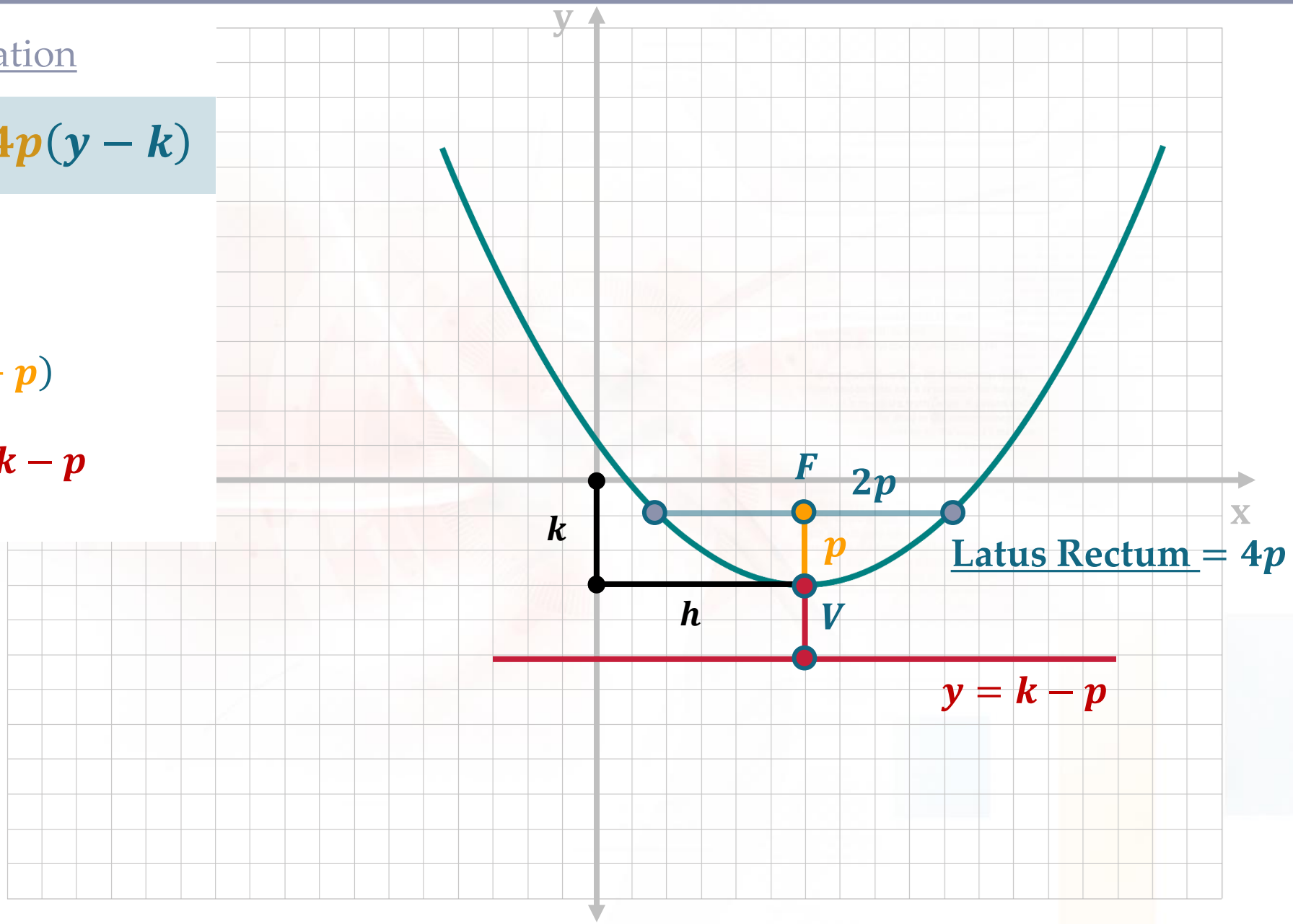
Standard Equation

$$(x-h)^2 = 4p(y-k)$$

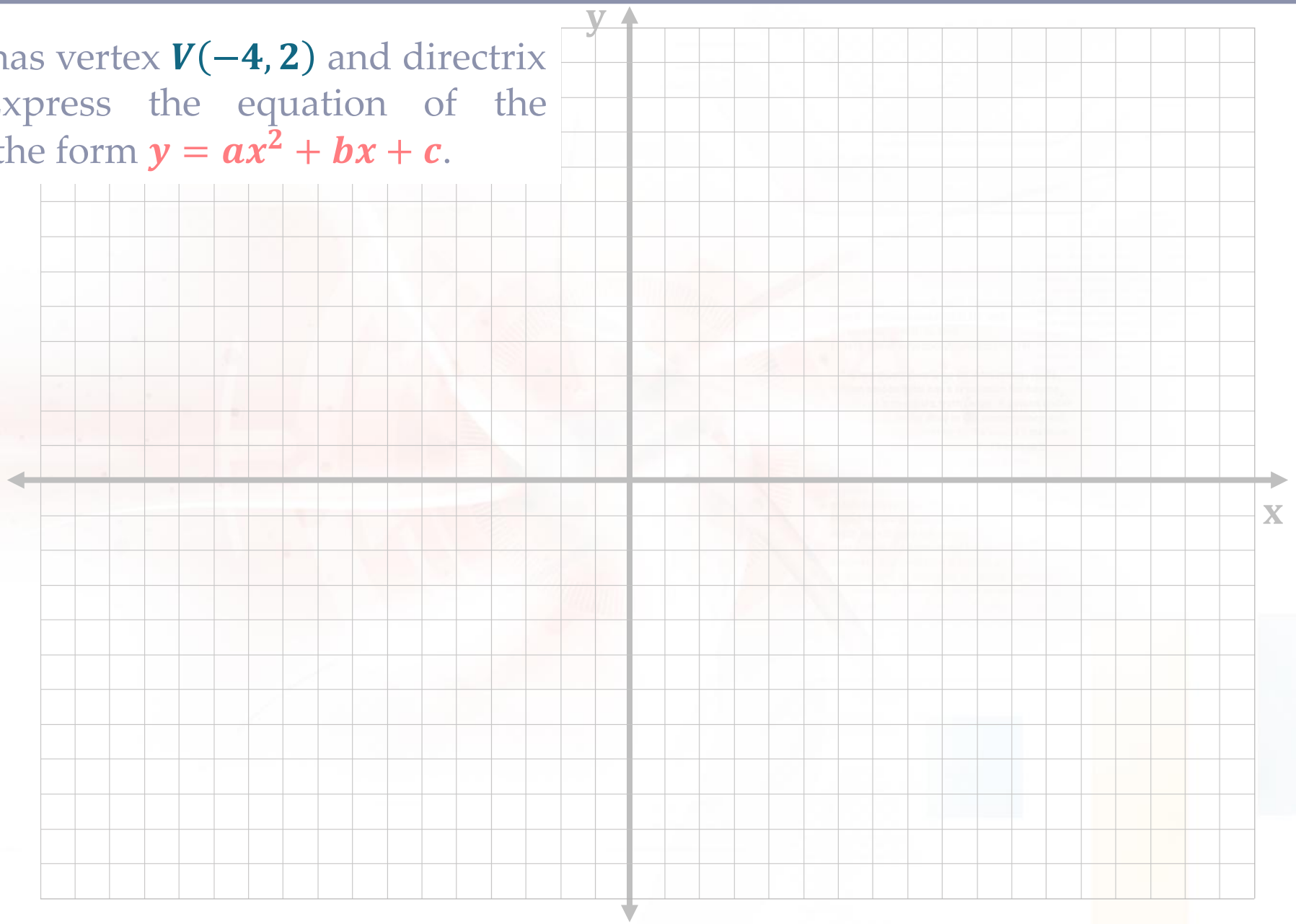
vertex, $V(h,k)$

focus, $F(h, k+p)$

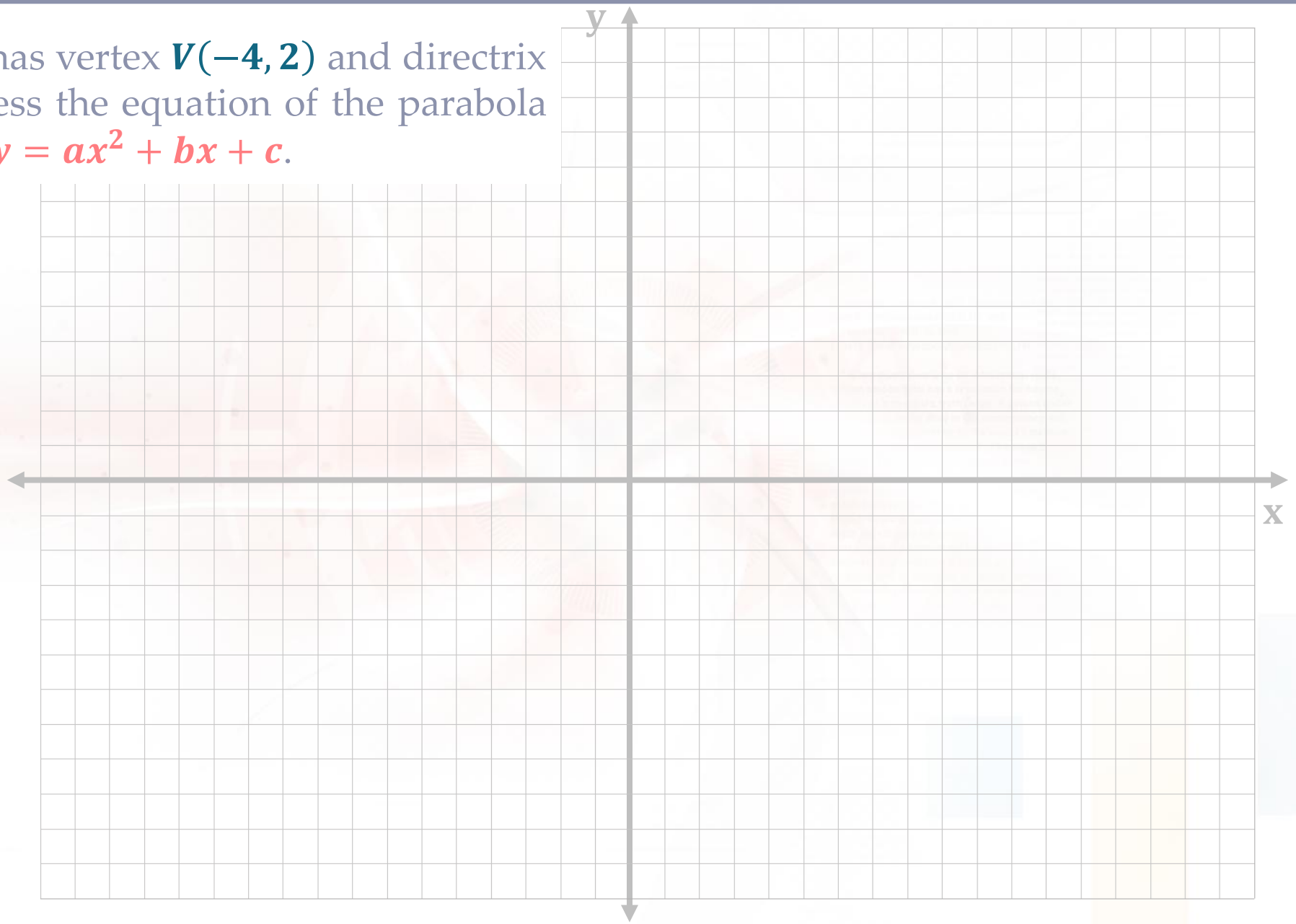
directrix, $y = k - p$



A parabola has vertex $V(-4, 2)$ and directrix $y = -5$. Express the equation of the parabola in the form $y = ax^2 + bx + c$.



A parabola has vertex $V(-4, 2)$ and directrix $y = 5$. Express the equation of the parabola in the form $y = ax^2 + bx + c$.



DIRECTRIX is along the $y - axis$

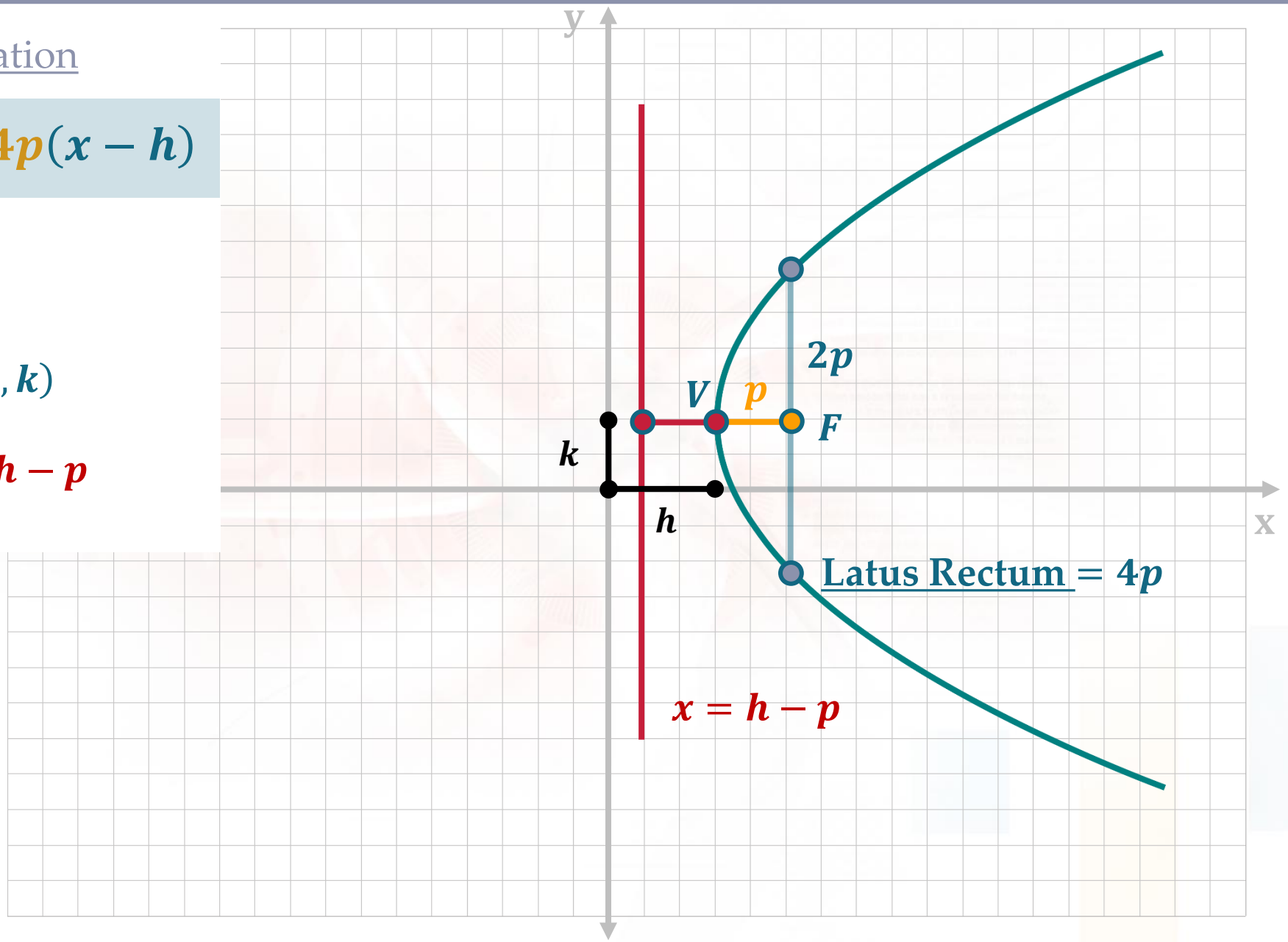
Standard Equation

$$(y-k)^2 = 4p(x-h)$$

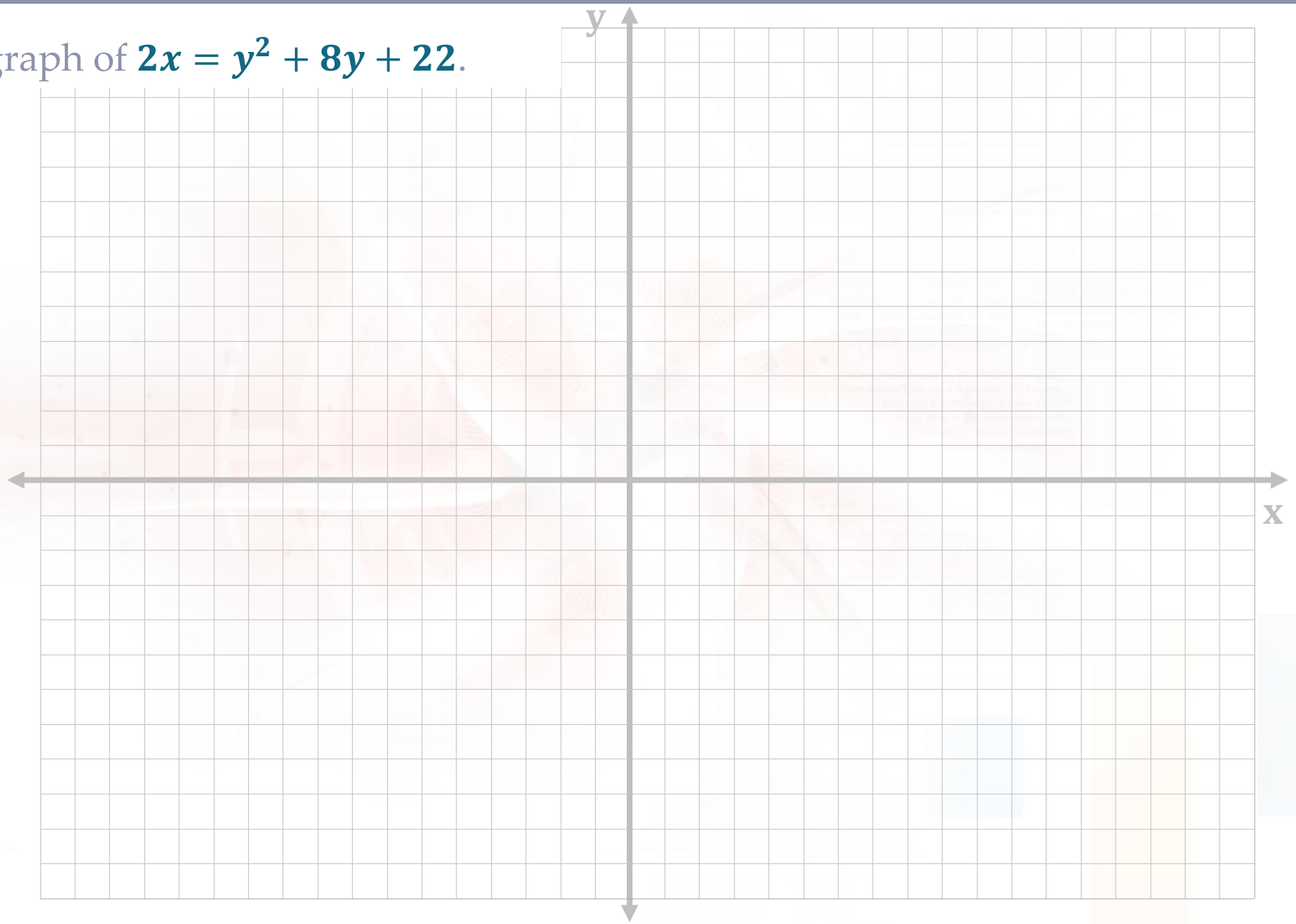
vertex, $V(h,k)$

focus, $F(h+p,k)$

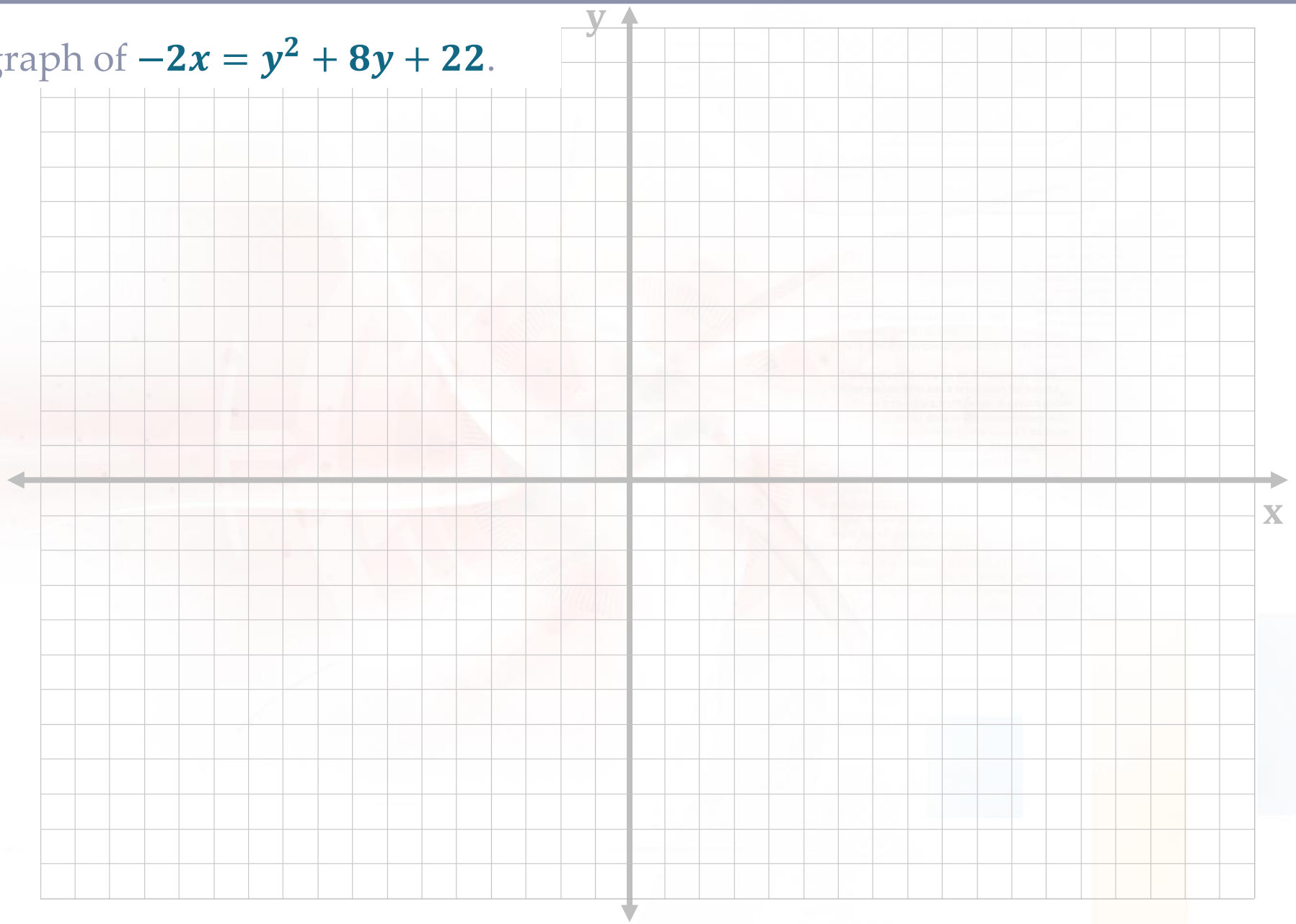
directrix, $x = h - p$



Sketch the graph of $2x = y^2 + 8y + 22$.



Sketch the graph of $-2x = y^2 + 8y + 22$.



SEATWORK